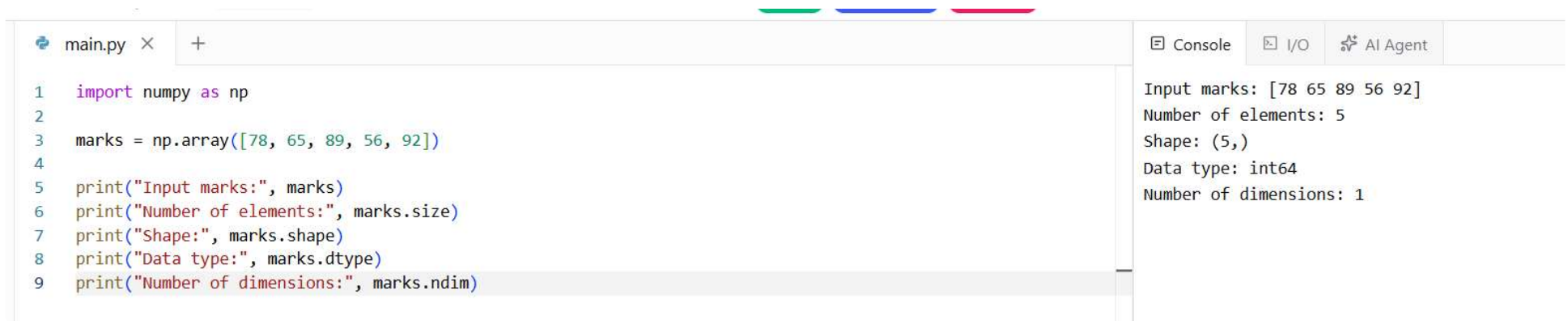


Question 1 – Student Marks Array

The marks obtained by five students in a subject are given as [78, 65, 89, 56, 92]. Create a NumPy array and display the array along with its basic properties.

```
import numpy as np
marks = np.array([78, 65, 89, 56, 92])
print("Input marks:", marks)
print("Number of elements:", marks.size)
print("Shape:", marks.shape)
print("Data type:", marks.dtype)
print("Number of dimensions:", marks.ndim)
```

output:



```
main.py × +
1 import numpy as np
2
3 marks = np.array([78, 65, 89, 56, 92])
4
5 print("Input marks:", marks)
6 print("Number of elements:", marks.size)
7 print("Shape:", marks.shape)
8 print("Data type:", marks.dtype)
9 print("Number of dimensions:", marks.ndim)
```

Console I/O AI Agent

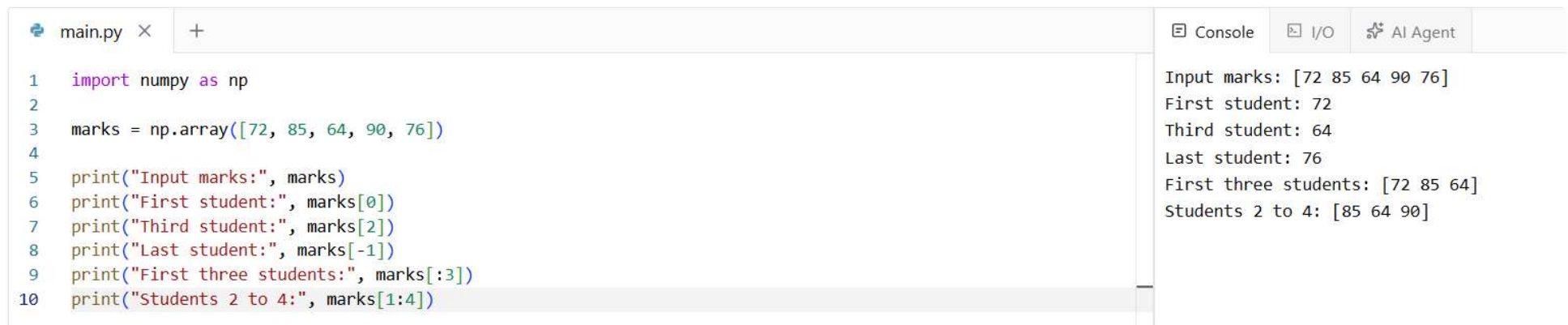
Input marks: [78 65 89 56 92]
Number of elements: 5
Shape: (5,)
Data type: int64
Number of dimensions: 1

Question 2 – Student Marks Access

The marks of five students are stored in a NumPy array as [72, 85, 64, 90, 76]. Write a program to access and display specific student marks using NumPy indexing and slicing.

```
import numpy as np
marks = np.array([72, 85, 64, 90, 76])
print("Input marks:", marks)
print("First student:", marks[0])
print("Third student:", marks[2])
print("Last student:", marks[-1])
print("First three students:", marks[:3])
print("Students 2 to 4:", marks[1:4])
```

output:



The screenshot shows a code editor with a file named 'main.py'. The code defines a NumPy array 'marks' with values [72, 85, 64, 90, 76] and prints several specific elements and slices. To the right, a 'Console' window displays the output of the program.

```
main.py × +
1 import numpy as np
2
3 marks = np.array([72, 85, 64, 90, 76])
4
5 print("Input marks:", marks)
6 print("First student:", marks[0])
7 print("Third student:", marks[2])
8 print("Last student:", marks[-1])
9 print("First three students:", marks[:3])
10 print("Students 2 to 4:", marks[1:4])
```

Console

```
Input marks: [72 85 64 90 76]
First student: 72
Third student: 64
Last student: 76
First three students: [72 85 64]
Students 2 to 4: [85 64 90]
```

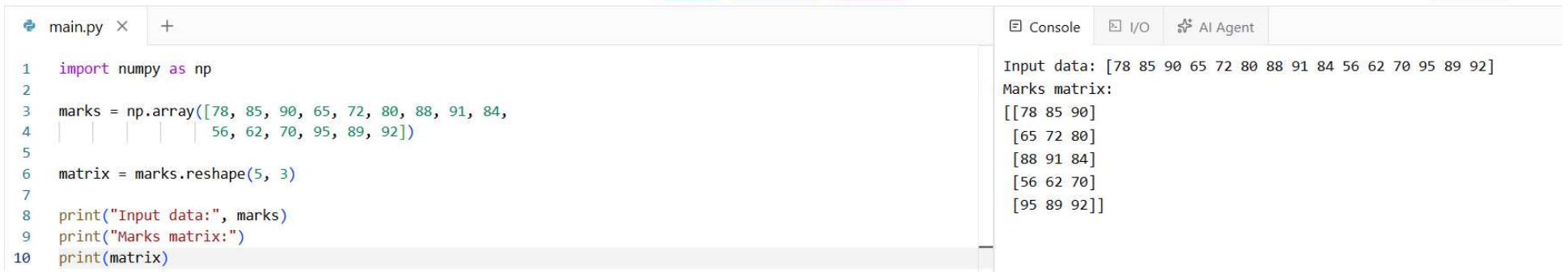
Question 3 – Subject-wise Marks

The marks obtained by five students in three subjects are given below.

Create a NumPy array to represent the data and reshape it into an appropriate matrix format.[78, 85, 90, 65, 72, 80, 88, 91, 84, 56, 62, 70, 95, 89, 92]

```
import numpy as np
marks = np.array([78, 85, 90, 65, 72, 80, 88, 91, 84,
                  56, 62, 70, 95, 89, 92])
matrix = marks.reshape(5, 3)
print("Input data:", marks)
print("Marks matrix:")
print(matrix)
```

output:



```
main.py x +
1 import numpy as np
2
3 marks = np.array([78, 85, 90, 65, 72, 80, 88, 91, 84,
4                  56, 62, 70, 95, 89, 92])
5
6 matrix = marks.reshape(5, 3)
7
8 print("Input data:", marks)
9 print("Marks matrix:")
10 print(matrix)
```

Console I/O AI Agent

Input data: [78 85 90 65 72 80 88 91 84 56 62 70 95 89 92]
Marks matrix:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]

Question 4 – Internal and External Marks

The internal and external examination marks of five students are stored in two NumPy arrays. Write a program to calculate the final marks of each student using NumPy array operations.

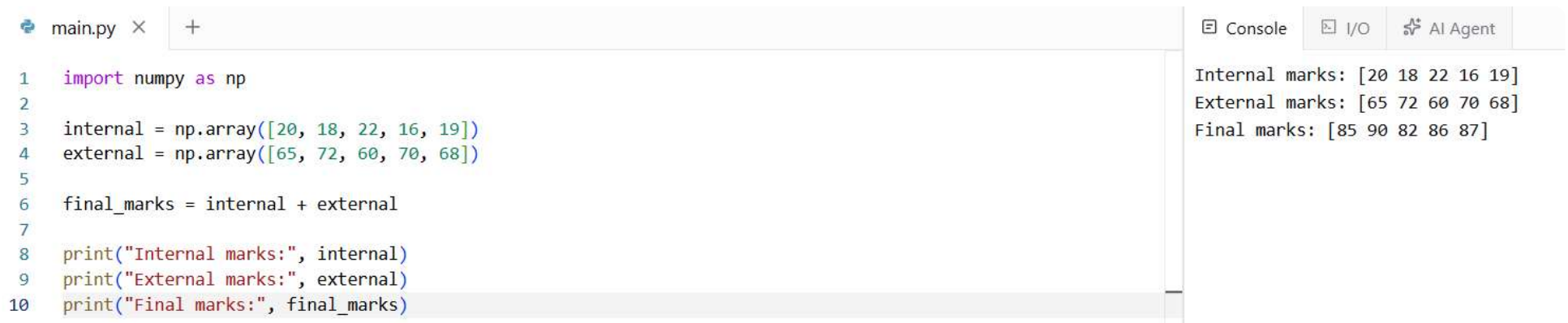
```
import numpy as np

internal = np.array([20, 18, 22, 16, 19])
external = np.array([65, 72, 60, 70, 68])

final_marks = internal + external

print("Internal marks:", internal)
print("External marks:", external)
print("Final marks:", final_marks)
```

output:



The screenshot shows a code editor with a file named 'main.py'. The code defines two NumPy arrays, 'internal' and 'external', and calculates their sum into 'final_marks'. The output is displayed in a console window on the right.

```
main.py × +
```

```
1 import numpy as np
2
3 internal = np.array([20, 18, 22, 16, 19])
4 external = np.array([65, 72, 60, 70, 68])
5
6 final_marks = internal + external
7
8 print("Internal marks:", internal)
9 print("External marks:", external)
10 print("Final marks:", final_marks)
```

Console I/O AI Agent

Internal marks: [20 18 22 16 19]
External marks: [65 72 60 70 68]
Final marks: [85 90 82 86 87]

Question 5 – Pass Percentage Analysis

The marks obtained by five students are [45, 78, 56, 32, 91]. Using NumPy Boolean masking, identify the students who have secured 50 marks or above.

```
import numpy as np

marks = np.array([45, 78, 56, 32, 91])

passed = marks[marks >= 50]

print("Input marks:", marks)
print("Marks 50 or above:", passed)
print("Number of students:", passed.size)
```

output:

```
main.py x +  
1 import numpy as np  
2  
3 marks = np.array([45, 78, 56, 32, 91])  
4  
5 passed = marks[marks >= 50]  
6  
7 print("Input marks:", marks)  
8 print("Marks 50 or above:", passed)  
9 print("Number of students:", passed.size)
```

Console I/O AI Agent

Input marks: [45 78 56 32 91]
Marks 50 or above: [78 56 91]
Number of students: 3

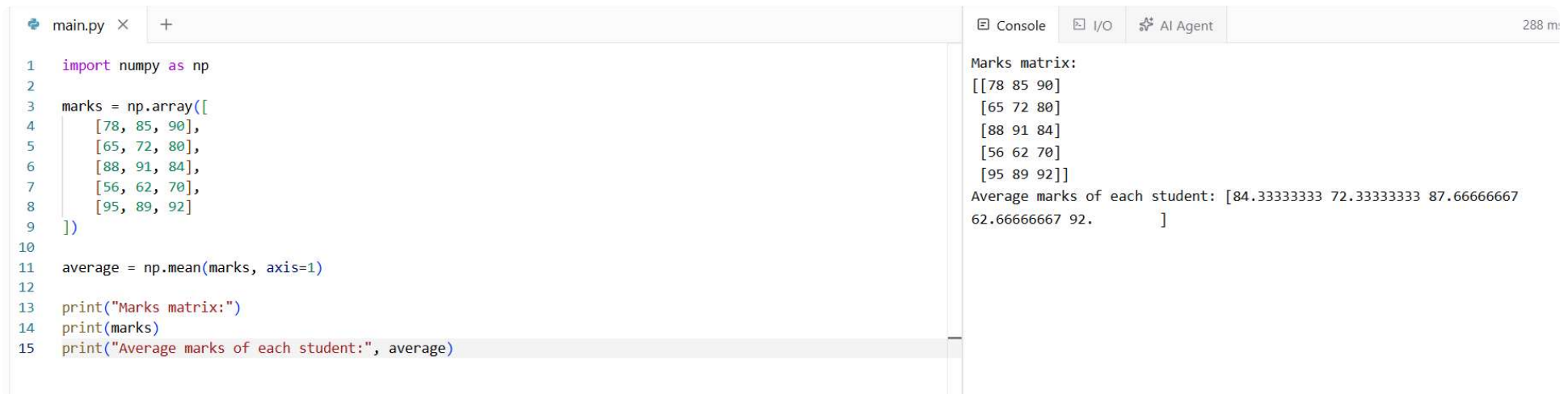
Question 6 – Average Marks

The marks of five students in three subjects are represented using a NumPy matrix. Write a program to calculate the average marks of each student.

```
import numpy as np  
  
marks = np.array([  
    [78, 85, 90],  
    [65, 72, 80],  
    [88, 91, 84],  
    [56, 62, 70],  
    [95, 89, 92]  
)  
  
average = np.mean(marks, axis=1)
```

```
print("Marks matrix:")
print(marks)
print("Average marks of each student:", average)
```

output:



The screenshot shows a code editor with a file named 'main.py'. The code defines a NumPy array 'marks' with 5 rows and 3 columns, representing marks for 5 students across 3 subjects. It then calculates the average marks for each student using 'np.mean' along axis 1. The output in the console shows the 'Marks matrix' as a 5x3 array and the 'Average marks of each student' as a 1D array with 5 elements.

```
1 import numpy as np
2
3 marks = np.array([
4     [78, 85, 90],
5     [65, 72, 80],
6     [88, 91, 84],
7     [56, 62, 70],
8     [95, 89, 92]
9 ])
10
11 average = np.mean(marks, axis=1)
12
13 print("Marks matrix:")
14 print(marks)
15 print("Average marks of each student:", average)
```

Console Output:

```
Marks matrix:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]
Average marks of each student: [84.33333333 72.33333333 87.66666667
 62.66666667 92.        ]
```

Question 7 – Class Performance Statistics

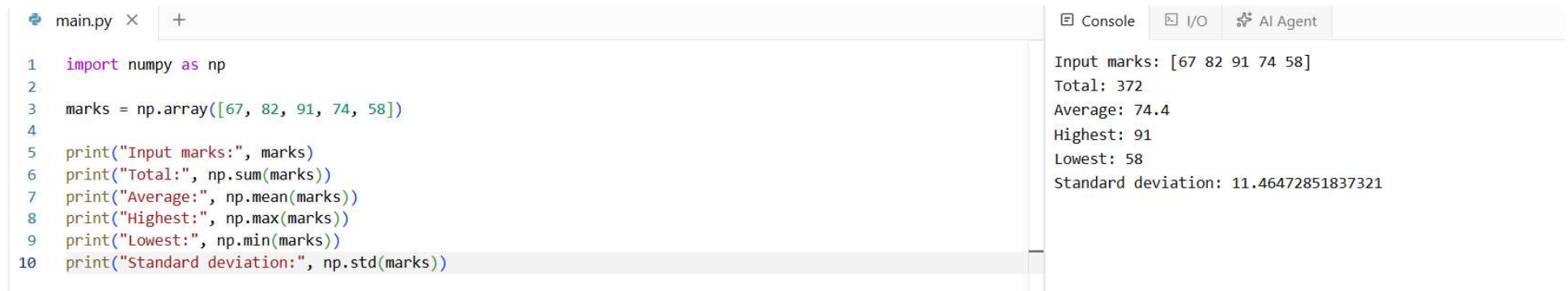
The marks obtained by five students are [67, 82, 91, 74, 58]. Using NumPy statistical functions, determine the total, average, highest, lowest, and standard deviation of the marks.

```
import numpy as np
```

```
marks = np.array([67, 82, 91, 74, 58])

print("Input marks:", marks)
print("Total:", np.sum(marks))
print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))
print("Standard deviation:", np.std(marks))
```

output:



The screenshot shows a code editor with a file named 'main.py'. The code defines a NumPy array 'marks' with values [67, 82, 91, 74, 58] and prints various statistics. To the right, a 'Console' panel displays the output of the code.

```
main.py × +
```

```
1 import numpy as np
2
3 marks = np.array([67, 82, 91, 74, 58])
4
5 print("Input marks:", marks)
6 print("Total:", np.sum(marks))
7 print("Average:", np.mean(marks))
8 print("Highest:", np.max(marks))
9 print("Lowest:", np.min(marks))
10 print("Standard deviation:", np.std(marks))
```

Console I/O AI Agent

Input marks: [67 82 91 74 58]
Total: 372
Average: 74.4
Highest: 91
Lowest: 58
Standard deviation: 11.46472851837321

Question 8 – Subject-wise Performance

The marks of five students in three subjects are stored in a NumPy matrix. Write a program to calculate the total marks obtained in each subject using an appropriate axis operation.

```
import numpy as np

marks = np.array([
    [78, 85, 90],
    [65, 72, 80],
    [88, 91, 84],
    [56, 62, 70],
    [95, 89, 92]
])

subject_total = np.sum(marks, axis=0)

print("Marks matrix:")
print(marks)
print("Total marks in each subject:", subject_total)
```

output:

```
main.py x +
1 import numpy as np
2
3 marks = np.array([
4     [78, 85, 90],
5     [65, 72, 80],
6     [88, 91, 84],
7     [56, 62, 70],
8     [95, 89, 92]
9 ])
10
11 subject_total = np.sum(marks, axis=0)
12
13 print("Marks matrix:")
14 print(marks)
15 print("Total marks in each subject:", subject_total)
```

Console I/O AI Agent

Marks matrix:
[[78 85 90]
[65 72 80]
[88 91 84]
[56 62 70]
[95 89 92]]
Total marks in each subject: [382 399 416]

Question 9 – Student-wise Performance

The marks of five students in three subjects are stored in a NumPy matrix. Write a program to calculate the total marks obtained by each student using an appropriate axis operation.

```
import numpy as np
```

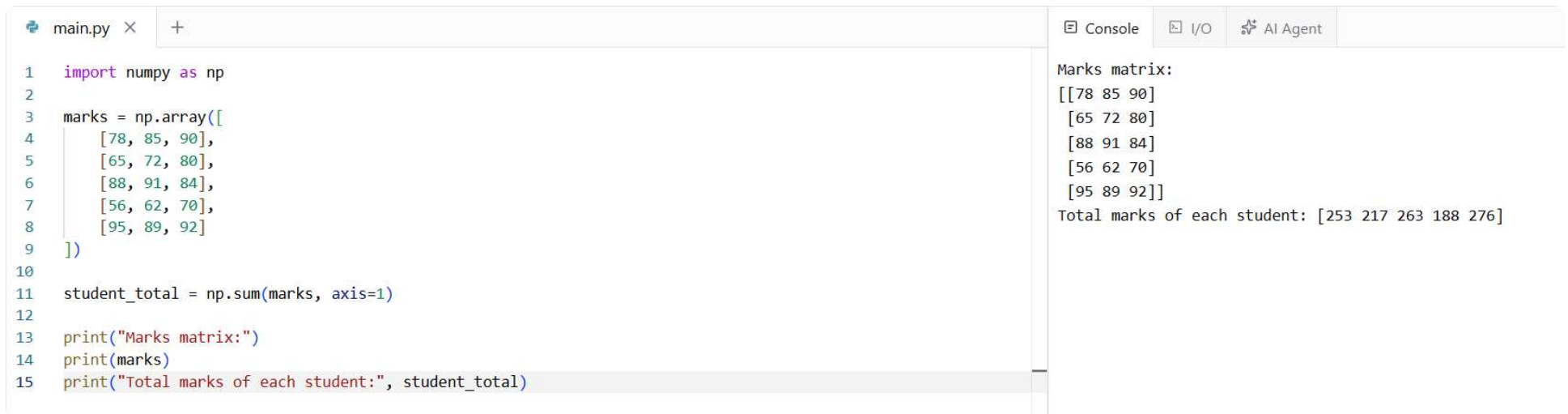
```
marks = np.array([
    [78, 85, 90],
    [65, 72, 80],
    [88, 91, 84],
    [56, 62, 70],
    [95, 89, 92]
```

```
] )

student_total = np.sum(marks, axis=1)

print("Marks matrix:")
print(marks)
print("Total marks of each student:", student_total)
```

output:



The screenshot shows a code editor with a file named 'main.py'. The code defines a NumPy array 'marks' with 5 rows and 3 columns, calculates the sum of marks for each student along axis 1, and prints the results. The output on the right shows the 'Marks matrix' as a 5x3 array and the 'Total marks of each student' as a 1D array.

```
main.py x +
```

```
1 import numpy as np
2
3 marks = np.array([
4     [78, 85, 90],
5     [65, 72, 80],
6     [88, 91, 84],
7     [56, 62, 70],
8     [95, 89, 92]
9 ])
10
11 student_total = np.sum(marks, axis=1)
12
13 print("Marks matrix:")
14 print(marks)
15 print("Total marks of each student:", student_total)
```

Console I/O AI Agent

Marks matrix:
[[78 85 90]
[65 72 80]
[88 91 84]
[56 62 70]
[95 89 92]]

Total marks of each student: [253 217 263 188 276]

Question 10 – Student Ranking

The total marks obtained by five students are [245, 278, 219, 290, 256]. Use NumPy sorting and indexing operations to arrange the marks in order and

determine the ranking of the students.

```
import numpy as np

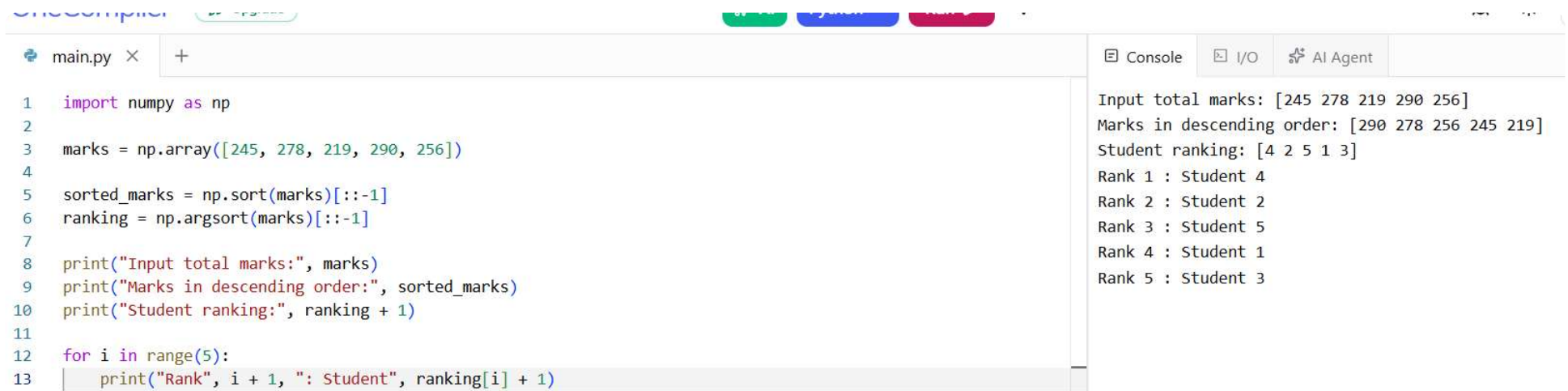
marks = np.array([245, 278, 219, 290, 256])

sorted_marks = np.sort(marks)[::-1]
ranking = np.argsort(marks)[::-1]

print("Input total marks:", marks)
print("Marks in descending order:", sorted_marks)
print("Student ranking:", ranking + 1)

for i in range(5):
    print("Rank", i + 1, ": Student", ranking[i] + 1)
```

output:



The screenshot shows a code editor with a file named `main.py`. The code uses NumPy to process a list of marks. The output is displayed in a console on the right side of the editor.

```
1 import numpy as np
2
3 marks = np.array([245, 278, 219, 290, 256])
4
5 sorted_marks = np.sort(marks)[::-1]
6 ranking = np.argsort(marks)[::-1]
7
8 print("Input total marks:", marks)
9 print("Marks in descending order:", sorted_marks)
10 print("Student ranking:", ranking + 1)
11
12 for i in range(5):
13     print("Rank", i + 1, ": Student", ranking[i] + 1)
```

Console Output:

```
Input total marks: [245 278 219 290 256]
Marks in descending order: [290 278 256 245 219]
Student ranking: [4 2 5 1 3]
Rank 1 : Student 4
Rank 2 : Student 2
Rank 3 : Student 5
Rank 4 : Student 1
Rank 5 : Student 3
```

Question 11 – Duplicate Marks Analysis

The marks obtained by five students are [85, 92, 85, 76, 92]. Use NumPy functions to identify the unique marks obtained by the students.

```
import numpy as np

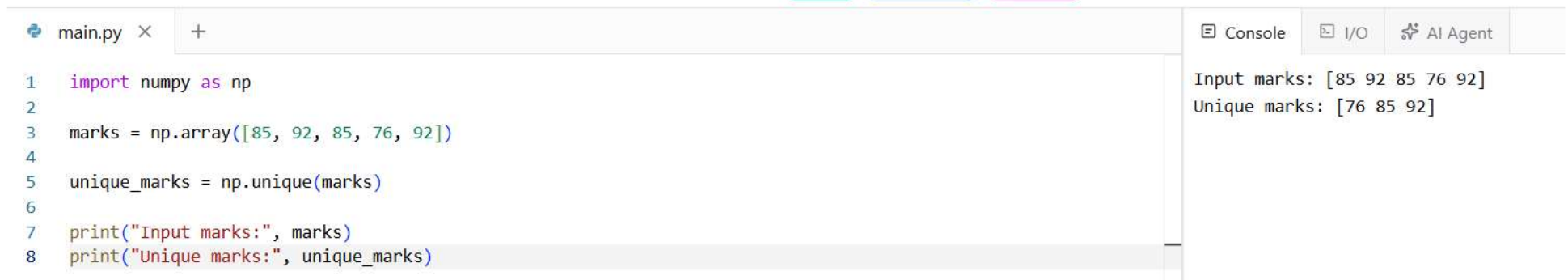
marks = np.array([85, 92, 85, 76, 92])

unique_marks = np.unique(marks)

print("Input marks:", marks)
```

```
print("Unique marks:", unique_marks)
```

output:



The screenshot shows a code editor with a file named 'main.py'. The code in the editor is as follows:

```
1 import numpy as np
2
3 marks = np.array([85, 92, 85, 76, 92])
4
5 unique_marks = np.unique(marks)
6
7 print("Input marks:", marks)
8 print("Unique marks:", unique_marks)
```

On the right side of the editor, there are tabs for 'Console', 'I/O', and 'AI Agent'. The 'Console' tab is active, displaying the output of the script:

```
Input marks: [85 92 85 76 92]
Unique marks: [76 85 92]
```

Question 12 – Missing Marks

The marks of five students are represented as `[78, 85, np.nan, 92, 67]`, where `np.nan` represents a missing mark. Write a NumPy program to calculate the average marks without considering the missing value.

```
import numpy as np

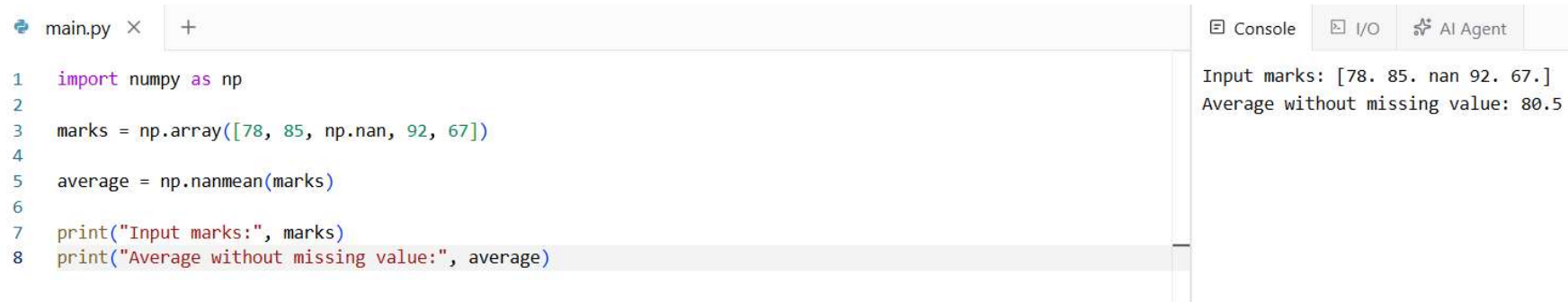
marks = np.array([78, 85, np.nan, 92, 67])

average = np.nanmean(marks)

print("Input marks:", marks)
```

```
print("Average without missing value:", average)
```

output:



The screenshot shows a code editor with a file named 'main.py'. The code defines a NumPy array 'marks' with values [78, 85, np.nan, 92, 67] and calculates its nanmean, which is 80.5. The output console on the right displays the input marks and the calculated average.

```
main.py × +  
1 import numpy as np  
2  
3 marks = np.array([78, 85, np.nan, 92, 67])  
4  
5 average = np.nanmean(marks)  
6  
7 print("Input marks:", marks)  
8 print("Average without missing value:", average)
```

Console I/O AI Agent
Input marks: [78. 85. nan 92. 67.]
Average without missing value: 80.5

Question 13 – Grade Classification

The marks obtained by five students are [95, 82, 74, 61, 45]. Using NumPy conditional operations, classify the students into appropriate grade categories based on their marks.

```
import numpy as np  
  
marks = np.array([95, 82, 74, 61, 45])  
  
grades = np.select(  
    [  

```

```
marks >= 90,  
marks >= 80,  
marks >= 70,  
marks >= 60  
],  
[  
    "A",  
    "B",  
    "C",  
    "D"  
],  
default="F"  
)  
  
print("Input marks:", marks)  
print("Grades:", grades)
```

output:

```
main.py x +  
1 import numpy as np  
2  
3 marks = np.array([95, 82, 74, 61, 45])  
4  
5 grades = np.select(  
6     [  
7         marks >= 90,  
8         marks >= 80,  
9         marks >= 70,  
10        marks >= 60  
11    ],  
12    [  
13        "A",  
14        "B",  
15        "C",  
16        "D"  
17    ],  
18    default="F"  
19 )  
20  
21 print("Input marks:", marks)  
22 print("Grades:", grades)
```

Console I/O AI Agent

Input marks: [95 82 74 61 45]
Grades: ['A' 'B' 'C' 'D' 'F']

Question 14 – Random Marks Generation

Generate marks for five students using NumPy's random number generation functionality. Perform basic statistical analysis on the generated marks.

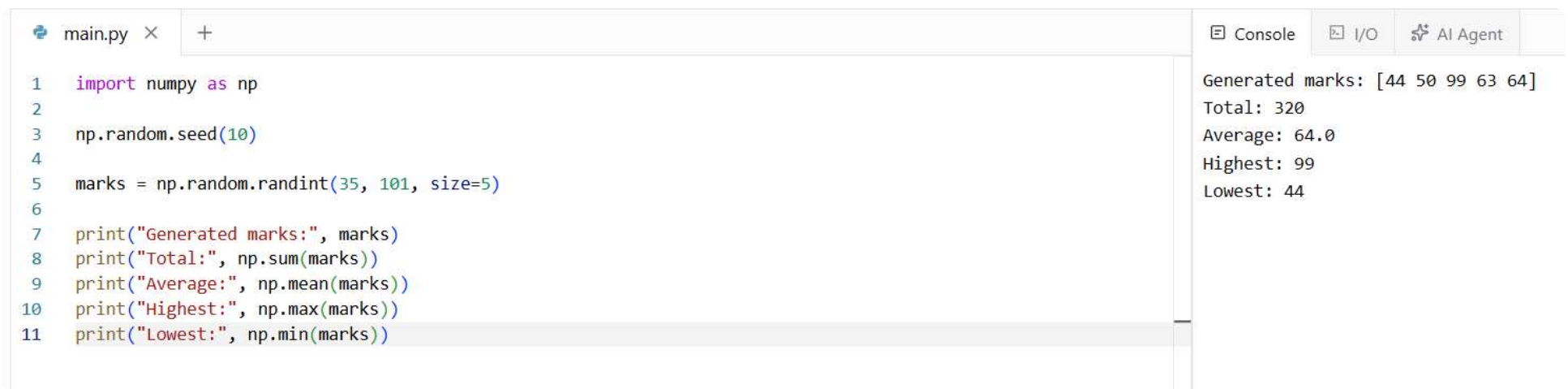
```
import numpy as np
```

```
np.random.seed(10)
```

```
marks = np.random.randint(35, 101, size=5)

print("Generated marks:", marks)
print("Total:", np.sum(marks))
print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))
```

output:



The screenshot shows a code editor with a file named 'main.py'. The code in the editor is as follows:

```
1 import numpy as np
2
3 np.random.seed(10)
4
5 marks = np.random.randint(35, 101, size=5)
6
7 print("Generated marks:", marks)
8 print("Total:", np.sum(marks))
9 print("Average:", np.mean(marks))
10 print("Highest:", np.max(marks))
11 print("Lowest:", np.min(marks))
```

On the right side of the editor, there is a 'Console' tab. The output displayed in the console is:

```
Generated marks: [44 50 99 63 64]
Total: 320
Average: 64.0
Highest: 99
Lowest: 44
```

Question 15 – Student Performance Analysis

The marks of five students in three subjects are stored in a NumPy array. Develop a program to perform a complete student performance analysis by

calculating the total marks, average marks, highest marks, lowest marks, and identifying students who perform above the class average.

```
import numpy as np
marks = np.array([
    [78, 85, 90],
    [65, 72, 80],
    [88, 91, 84],
    [56, 62, 70],
    [95, 89, 92]
])
total = np.sum(marks, axis=1)
average = np.mean(marks, axis=1)
highest = np.max(marks, axis=1)
lowest = np.min(marks, axis=1)
class_average = np.mean(total)
above_average = np.where(total > class_average)[0] + 1
print("Marks matrix:")
print(marks)
print("Total marks:", total)
print("Average marks:", average)
print("Highest mark:", highest)
print("Lowest mark:", lowest)
print("Class average total:", class_average)
print("Students above class average:", above_average)
```

output:

```
main.py × +  
1 import numpy as np  
2  
3 marks = np.array([  
4     [78, 85, 90],  
5     [65, 72, 80],  
6     [88, 91, 84],  
7     [56, 62, 70],  
8     [95, 89, 92]  
9 ])  
10  
11 total = np.sum(marks, axis=1)  
12 average = np.mean(marks, axis=1)  
13 highest = np.max(marks, axis=1)  
14 lowest = np.min(marks, axis=1)  
15  
16 class_average = np.mean(total)  
17  
18 above_average = np.where(total > class_average)[0] + 1  
19  
20 print("Marks matrix:")  
21 print(marks)  
22  
23 print("Total marks:", total)  
24 print("Average marks:", average)  
25 print("Highest mark:", highest)  
26 print("Lowest mark:", lowest)  
27 print("Class average total:", class_average)  
28 print("Students above class average:", above_average)
```

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Marks matrix:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]
Total marks: [253 217 263 188 276]
Average marks: [84.33333333 72.33333333 87.66666667 62.66666667 92.
]
Highest mark: [90 80 91 70 95]
Lowest mark: [78 65 84 56 89]
Class average total: 239.4
Students above class average: [1 3 5]